JAKE A. SOLOFF

Website: jake-soloff.github.io ⋄ Email: soloff@umich.edu

ACADEMIC APPOINTMENTS

Assistant Professor, *University of Michigan* 7/2025 – Present

Postdoctoral Scholar, *The University of Chicago* 6/2022 – 6/2025
Advisors: Rina Foygel Barber and Rebecca Willett

EDUCATION

PhD in Statistics, *University of California, Berkeley* 8/2016 – 5/2022
Advisors: Adityanand Guntuboyina and Michael I. Jordan

ScB in Mathematics, *Brown University* 9/2012 – 5/2016
Advisors: Richard Evan Schwartz and Erik B. Sudderth

PUBLICATIONS

Soloff, J. A., Barber, R. F., & Willett, R. (2026). Stability via resampling: statistical problems beyond the real line. *To appear in the Electronic Journal of Statistics.* [preprint]

Bates, S., Jordan, M. I., Sklar, M., & Soloff, J. A. (2026). Principal-agent hypothesis testing. *To appear in the Journal of the American Statistical Association.* [paper] [preprint]

Liu, S., Panigrahi, S., & Soloff, J. A. (2026). Cross-validation with antithetic Gaussian randomization. *Journal of the Royal Statistical Society: Series B, qkag073.* [paper] [preprint]

Xiang, D., Soloff, J. A., & Fithian, W. (2026). A frequentist local false discovery rate. *Biometrika, 113(1).* [paper] [preprint]

Adrian, M., Soloff, J. A., & Willett, R. (2025). Stabilizing black-box model selection with the inflated argmax. *Transactions on Machine Learning Research (TMLR).* [paper] [preprint]

Rossellini, R., Soloff, J. A., Barber, R. F., Ren, Z., & Willett, R. (2025). Can a calibration metric be both testable and actionable? *38th Conference on Learning Theory (COLT 2025).* [paper] [preprint]

Soloff, J. A., Barber, R. F., & Willett, R. (2024). Building a stable classifier with the inflated argmax. *Advances in Neural Information Processing Systems 37 (NeurIPS 2024).* [paper] [preprint]

Soloff, J. A., Guntuboyina, A., & Sen, B. (2024). Multivariate, heteroscedastic empirical Bayes via nonparametric maximum likelihood. *Journal of the Royal Statistical Society: Series B, 87(1), 1-32.* [paper] [preprint]

Soloff, J. A., Barber, R. F., & Willett, R. (2024). Bagging provides assumption-free stability. *Journal of Machine Learning Research*, 25(131), 1-35. [paper] [preprint]

Soloff, J. A., Xiang, D., & Fithian, W. (2024). The edge of discovery: Controlling the local false discovery rate at the margin. *Annals of Statistics*, 52(2), 580-601. [paper] [preprint]

Soloff, J. A., Guntuboyina, A., & Pitman, J. (2019). Distribution-free properties of isotonic regression. *Electronic Journal of Statistics*, 13(2), 3243-3253. [paper] [preprint]

Soloff, J. A., Márquez, R. A., & Friedler, L. M. (2015). Products of geodesic graphs and the geodetic number of products. *Discussiones Mathematicae Graph Theory*, 35(1), 35–42. [paper]

PREPRINTS

Wadekar, A. S. & Soloff, J. A. (2026). Calibration without labels in multiple testing. [preprint]

Xiang, D., Fithian, W., Ignatiadis, N., Soloff, J. A., & Weinstein, A. (2026). Estimating the local false discovery rate under an unknown symmetric null. [preprint]

Liang, R., Soloff, J. A., Barber, R. F., & Willett, R. (2025). Assumption-free stability for ranking problems. [preprint]

Hore, R., Soloff, J. A., Barber, R. F., & Samworth, R. J. (2024). Testing conditional independence under isotonicity. [preprint]

Bates, S., Jordan, M. I., Sklar, M., & Soloff, J. A. (2023). Incentive-theoretic Bayesian inference for collaborative science. [preprint]

Soloff, J. A., Guntuboyina, A., & Jordan, M. I. (2020). Covariance estimation with nonnegative partial correlations. [preprint]

CONFERENCE ABSTRACTS

Shriver, J., Soloff, J. A., & Molen, N. (2014). Delivering the benefits of remotely sensed data and decision support tools to farmers. *American Geophysical Union.* [abstract]

SOFTWARE

`npeb`: Python package for nonparametric empirical Bayes methods.

SCHOLARSHIPS AND AWARDS

NSF Grant DMS-2610643, "Calibrated Hypothesis Testing"	2026
Eric Lehmann Citation for outstanding dissertation in theoretical statistics	2022
IMS Hannan Graduate Student Travel Award	2022
Bridgewater Fellowship in Data Science	2020
David Blackwell Fund	2019
Outstanding Graduate Student Instructor Award	2018
First place team, Citadel National Data Open Championship	2017
First place team, Citadel Datathon at UC Berkeley	2017
Leonard Chung-Wei Cheng Graduate Student Fund in Statistics	2016
Albert A. Bennett Prize for exceptional accomplishment in mathematics major	2016
Jerome L. Stein Memorial Award for undergraduate excellence in applied math	2016
Phi Beta Kappa, elected junior year	2015
Sidney E. Frank Scholarship	2012 – 2016
Dean's Scholarship, Brown summer session	2011

PROFESSIONAL ACTIVITIES

Workshop Organizer

"Algorithmic stability: mathematical foundations for the modern era" at the American Institute of Mathematics

Devoted to building a foundational understanding of algorithmic stability and developing rigorous tools for measuring stability that can characterize the behavior of machine learning algorithms.

Invited Talks

[23]	ICSDS (session on Advances in selective inference and multiple testing)	Dec. 2026
[22]	Joint Statistical Meetings (sessions on distribution-free inference and resampling methods)	Aug. 2026
[21]	IMS Annual Meeting (session on Empirical Bayes for approximate oracle inference)	Jul. 2026
[20]	Wayne State University Data Science Seminar	Apr. 2026
[19]	Workshop on Uncertainty Quantification for Decision Making (at UC Berkeley)	Oct. 2025
[18]	University of Michigan Statistics Student Seminar	Sep. 2025
[17]	UIUC Statistics Seminar	Feb. 2025
[16]	University of Michigan Statistics Seminar	Jan. 2025
[15]	Loyola University Chicago Statistics Seminar	Jan. 2025
[14]	Illinois Tech Statistics Seminar	Jan. 2025

[13] Carnegie Mellon University Statistics Seminar Jan. 2025

[12] University of Michigan Statistics Student Seminar Oct. 2023

[11] Wayne State University Data Science Seminar Oct. 2023

[10] Indian International Statistical Association (IISA) Conference Jun. 2023

[9] Notre Dame Statistics Seminar May 2023

[8] Matthew Stephens Lab, University of Chicago Apr. 2023

[7] Systems, Information, Learning, Optimization (SILO) Seminar, UW Madison [recording] Mar. 2023

[6] University of Bristol Statistics Seminar Nov. 2022

[5] 12th International Conference on Multiple Comparison Procedures Aug. 2022

[4] Discussant, International Seminar on Selective Inference [recording] Jul. 2022

[3] International Seminar on Selective Inference [recording] Mar. 2022

[2] Fourth Annual Berkeley-Stanford Econometrics Jamboree Nov. 2021

[1] Berkeley-Davis joint colloquium Apr. 2021

Referee Service

Annals of Applied Statistics (AoAS); Annals of Statistics (AoS); Biometrika; Conference on Learning Theory (COLT); Econometrica; Electronic Journal of Statistics (EJS); Journal of the American Statistical Association (JASA); Journal of Causal Inference (JCI); Journal of the Royal Statistical Society: Series B (JRSS-B); Operations Research (OR); Proceedings of the National Academy of Sciences (PNAS); Statistical Science

TEACHING EXPERIENCE

Instructor at University of Michigan 2025 – present

Advanced Statistical Computing (PhD level)

Applied Linear Regression

Graduate student instructor at UC Berkeley 2017 – 2020

Data, Inference, and Decisions, taught by M. I. Jordan and J. Steinhardt.

Theoretical Statistics II (PhD level), taught by M. I. Jordan.

Head GSI, Principles and Techniques of Data Science, taught by F. Perez and J. Gonzalez.

Principles and Techniques of Data Science, taught by D. Nolan and J. Gonzalez.

Assistant instructor at RI Department of Corrections 2016

Introductory Geography, taught by S. Bloch.

Teaching assistant at Brown University 2014 – 2015

Curricular Advising Fellow, Crime and the City, taught by S. Bloch.

Recent Applications of Probability and Statistics (PhD level), taught by S. Geman.

Computational Probability and Statistics, taught by S. Geman.

Crime and the City, taught by S. Bloch.